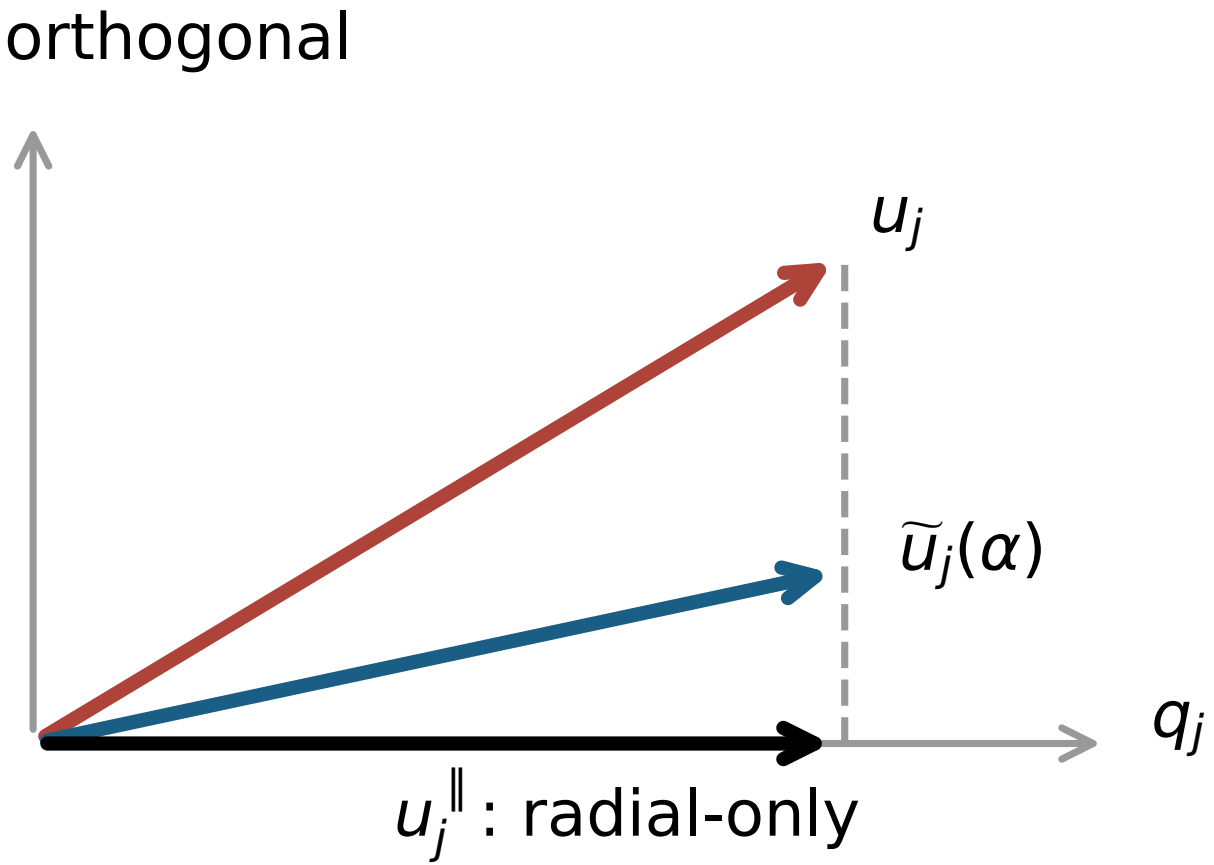
Figure 1



Reference-direction consolidation. The local update u_j is constrained toward the fixed reference q_j; alpha=0 retains only its parallel component. This conceptual sketch is not an empirical kernel projection.

Algorithm 1

At formation boundary:

$q \leftarrow \text{normalize}(w)$

For each subsequent update:

$u \leftarrow \text{LocalSTDP}(\text{spike traces})$

$u \leftarrow \alpha u + (1 - \alpha) q (q^{\mathsf{T}} u)$

$w \leftarrow \text{Regulate}(w + u)$

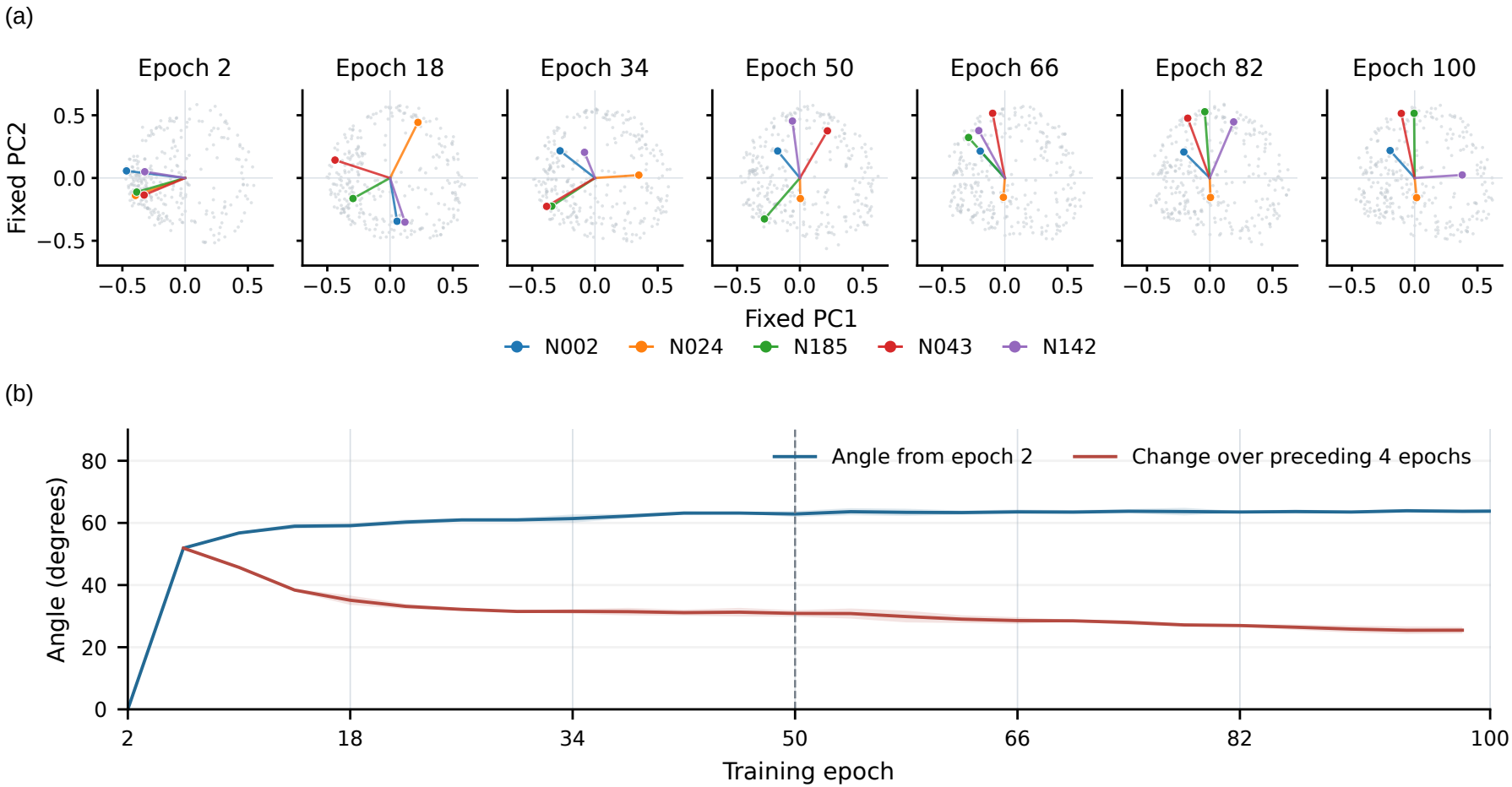
Reference-constrained hidden update

Table I

Setting	MNIST	N-MNIST
Input channels / time steps	6 / 40	2 / 20
Hidden channels	300	330
Hidden kernel / stride	10 x 10 / 2	9 x 9 / 2
Output neurons per class	20	45
Trace decay gamma	0.66	0.97
Lateral inhibition factor	3	30
Temporal margin	0.30	0.35
Positive / negative teaching gains	40 / 120	32 / 100
Group softmax temperature	0.25	0.25
Threshold bounds	[0.1, 10]	[1, 100]
Threshold log-step	0.00125	0.00125
Hidden amplifier log-step	6.25e-6	0
Output amplifier log-step	1.25e-6	0
Per-receiver normalization	L1	L1

Principal protocol settings. The output classifier has ten class groups. Hidden pooling uses 2 x 2 max pooling; output pooling is global.

Figure 2



Continued kernel-direction drift under local plasticity. (a) Five neurons uniformly sampled without replacement from the 300-neuron hidden layer of one MNIST run, tracked at epochs 2, 18, 34, 50, 66, 82, and 100. No rotation-angle, quadrant, or performance filter was used. Each incoming kernel is normalized to unit L_2 norm and projected into a single centered PCA basis fitted to all saved checkpoints of this run. The two displayed components explain 30.3% of centered variance. Neuron identities, colors, axes, and scale remain fixed; gray points show all 300 neurons. Spokes originate at the common PCA mean, not at the physical zero kernel, so their projected angles are not full-dimensional kernel angles. (b) Angles in the original 600-dimensional kernel space, relative to epoch 2 (blue) and to the preceding saved checkpoint four epochs earlier (red). Angles are averaged over all 300 neurons within each seed; curves and shaded bands show the mean and Student-t 95% confidence interval across four seeds. The terminal two-epoch interval is excluded from the red curve. The dashed line marks model-state continuation at epoch 50 without restored RNG or data-loader state; runs may use different GPU models. Nonzero local angular change can persist while distance from the early reference plateaus. This visualization describes drift and does not alone establish its causal effect on native decisions.

Table II

(A) Aggregate kernel statistics at each checkpoint

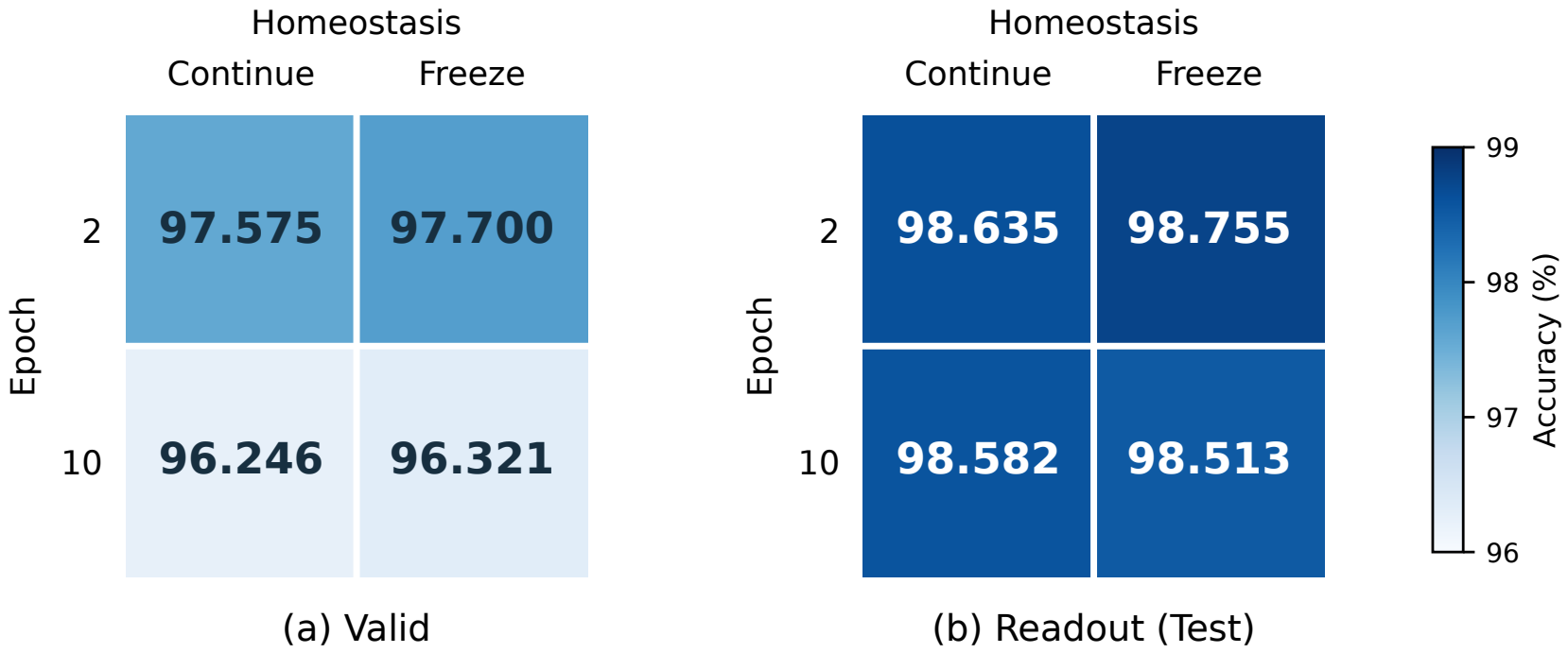
Statistic	Epoch 10	Epoch 30	Epoch 50	Epoch 70	Epoch 90
Participation ratio	4.90975	5.54409	5.78613	6.02215	5.94795
Global DC energy fraction	0.40941	0.37834	0.36742	0.36042	0.36432
Mean receiver DC fraction	0.48810	0.45970	0.44127	0.43116	0.43160
Signed mass	0.99768	0.99395	0.99248	0.99244	0.99419

(B) Directional change over consecutive 20-epoch intervals

Statistic	Baseline	10 → 30	30 → 50	50 → 70	70 → 90
Mean direction cosine	-	0.536	0.620	0.671	0.722
Mean direction angle (0-180°)	-	55.6°	47.5°	42.1°	36.0°
Orthogonal fraction (0-100%)	-	87.2%	88.8%	89.9%	90.9%

Nearly stable aggregate statistics coexist with continued directional drift (four seeds). (A) Mean kernel statistics at each checkpoint. (B) Directional changes over the labeled 20-epoch intervals, averaged over 300 matched receivers per seed, then across seeds. Orthogonal fraction is orthogonal displacement norm divided by total displacement norm, relative to the interval's starting direction, expressed as a percentage. It is not an energy or neuron-count fraction. The 50-70 interval crosses a model-state restart (see text).

Figure 3



Kernel boundary and homeostasis (four paired seeds). Rows indicate the epoch before which hidden-kernel learning is frozen; columns indicate continued or frozen hidden homeostasis. All conditions are evaluated at epoch 20. Cells show mean accuracy (%). Left: native validation (Valid). Right: hidden linear Readout test accuracy (Readout). Both panels share a truncated 96-99% scale. The panels use different data splits, not a paired validation-to-test comparison. This matched schedule comparison does not isolate pure rotation, and GPU models are not fully matched.

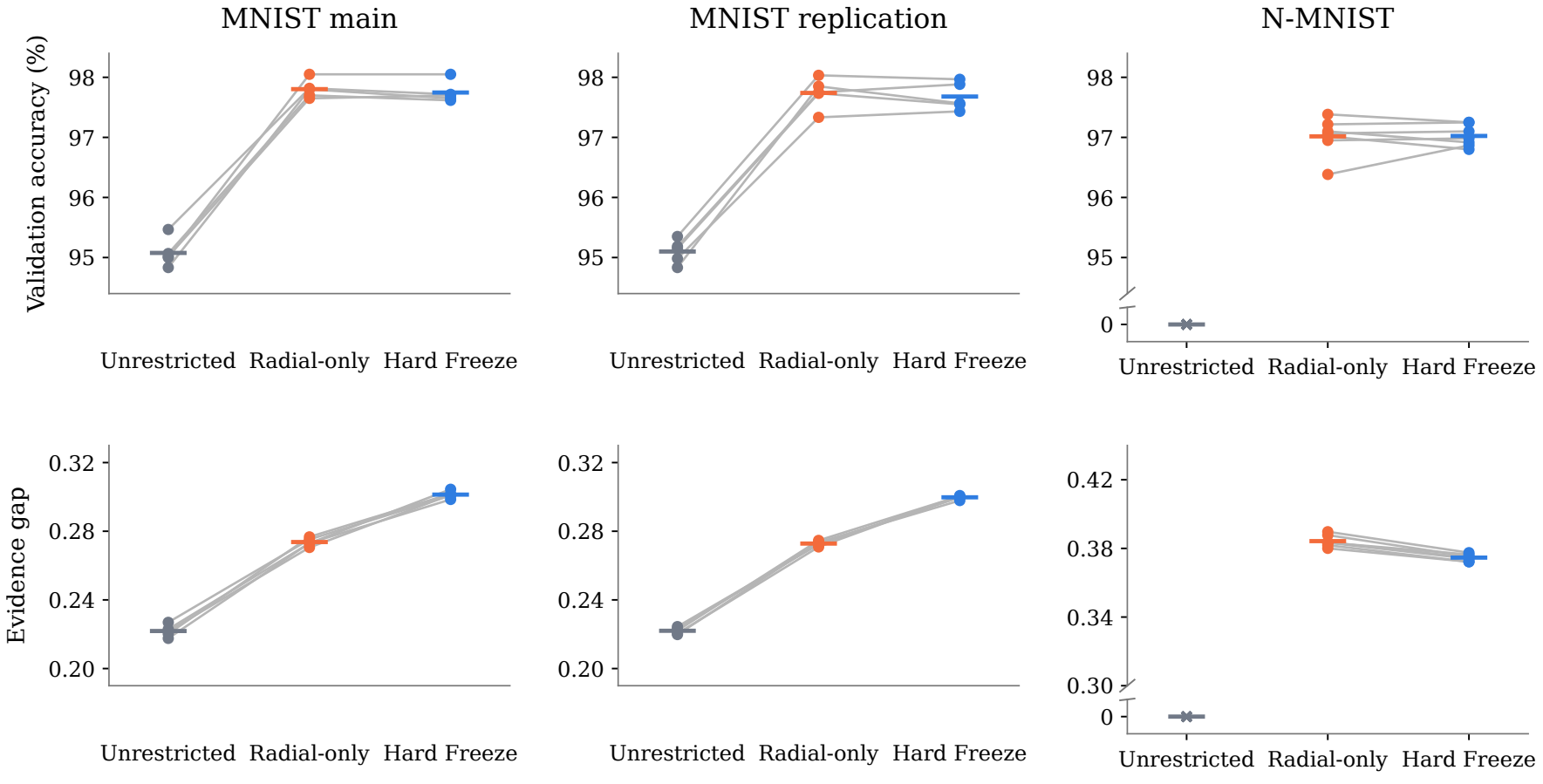
Table III

Dataset / batch	Condition	Seeds	Validation (%)	Test (%)
MNIST main	Unrestricted	5	95.077	95.484
MNIST main	Hard Freeze @2	5	97.747	98.008
MNIST main	Radial-only @2	5	97.803	97.954
MNIST replication	Unrestricted	5	95.100	95.590
MNIST replication	Hard Freeze @2	5	97.680	97.948
MNIST replication	Radial-only @2	5	97.740	97.832
N-MNIST	Unrestricted (100% no-decision)	7	0.000	0.000
N-MNIST	Hard Freeze @2	7	97.024	97.173
N-MNIST	Radial-only @2	7	97.017	97.284

Radial minus hard freeze	Validation diff. (pp)	Test diff. (pp)	Timing-gap difference
MNIST main	+0.057 [-0.043, 0.157]	-0.054 [-0.171, 0.063]	-0.02763 [-0.03151, -0.02375]
MNIST replication	+0.060 [-0.162, 0.282]	-0.116 [-0.297, 0.065]	-0.02695 [-0.02953, -0.02437]
N-MNIST	-0.007 [-0.226, 0.211]	+0.111 [0.026, 0.197]	+0.00961 [0.00771, 0.01151]

Final epoch-20 native accuracy and paired radial-only contrasts. Means are percentages; differences are percentage points with 95% paired Student-t CIs. Timing uses normalized earliness, not accuracy units. MNIST replication uses independent seeds.

Figure 4



Native validation accuracy and evidence gap at epoch 20. Conditions are unrestricted learning, radial-only updates, and hard freeze, from left to right. Gray lines connect matched seeds; colored points show individual runs and horizontal ticks show condition means. MNIST main and replication use the same training protocol with independent sets of five seeds; N-MNIST uses seven seeds. Top: native validation accuracy (%). Bottom: correct-minus-strongest-wrong earliness; larger gaps indicate greater temporal separation. The main axes span 4 percentage points for accuracy and 0.14 for gap; gap offsets differ between datasets. Broken N-MNIST axes retain the unrestricted zero values (crosses): all seven unrestricted runs have 100% no-decision. Their zero gap represents silence, not normal temporal decisions, and is not connected to active conditions. Hard freeze disables hidden learning and homeostasis; radial-only retains homeostasis. Seed pairing does not remove mixed-GPU confounding. Paired contrasts and CIs are given in Table III.